Are You Getting What You Pay For?
Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Audience: ML researchers, AI security practitioners, and cloud infrastructure engineers

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

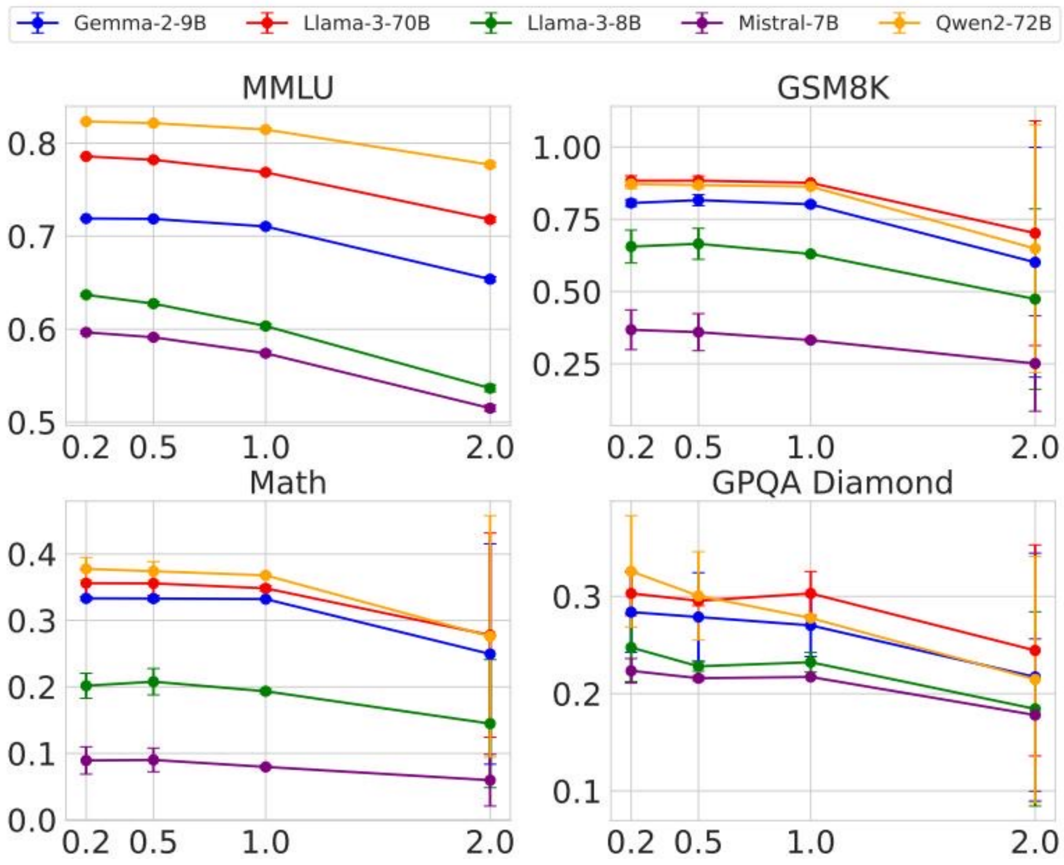
- Commercial LLM API providers may secretly replace advertised models to cut costs while maintaining pricing.
- Four Types of Covert Substitutions:
  - Smaller model (e.g., 8B instead of 70B)
  - Quantized variant (e.g., FP8/INT8 instead of FP16)
  - Fine-tuned or updated version
  - Entirely different model family
- Formal Hypothesis Testing Framework:
  - Null Hypothesis (H0): The API serves the specified model P_spec(y|x).
  - Alternative Hypothesis (H1): The API serves a substitute model P_actual(y|x).

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct (Orig vs FP8)	62.69 vs 62.43	61.14 vs 60.90	20.65 vs 14.91	22.62 vs 20.14
Llama-3-70B-Instruct (Orig vs FP8)	78.05 vs 77.88	88.06 vs 87.35	35.69 vs 35.75	29.60 vs 33.12
Gemma-2-9b-it (Orig vs FP8)	71.86 vs 71.92	81.80 vs 79.41	33.41 vs 32.53	28.64 vs 27.81
Qwen2-72B-Instruct (Orig vs FP8)	82.18 vs 81.98	86.72 vs 86.82	37.39 vs 37.67	29.93 vs 31.08
Mistral-7B-v0.3 (Orig vs FP8)	59.15 vs 58.77	35.90 vs 32.20	8.94 vs 7.68	21.60 vs 22.72

Key Insight: Performance drops from FP8 quantization are extremely small (often within error bars), especially on MMLU and GSM8K. This minimal benchmark drop motivates why substitution detection is highly difficult.

Higher Temperature Amplifies Performance Gaps but Increases Noise



At a temperature of 2.0, performance degrades across the board. The gap between original and substitute models narrows or becomes increasingly noisy, severely complicating statistical detection.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The same Llama-3-8B model produces varying log probabilities across different software frameworks (transformers vs. vLLM) and hardware (A100 vs. H100). This production-level nondeterminism fundamentally defeats metadata-based detection methods.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

- Adversarial Strategy: Randomized Substitution
  - Providers can mix substitute responses with a small probability (p).
  - As p approaches 0, the mixed distribution tightly converges to the originally specified model's distribution.
- Detection Becomes Impractical:
  - Because the distributions become nearly identical, an auditor requires exponentially more API queries to achieve statistical significance.
  - This makes independent detection economically and computationally prohibitive for users.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Fails against subtle substitutions (e.g., FP8).	Requires massive, costly query budgets to achieve sufficient statistical power.
Log-Probability Methods	Defeated by inference nondeterminism.	Cannot distinguish between legitimate infrastructure differences (hardware/software) and malicious swaps.
Adversarial Countermeasures	Benchmark evasion & Randomized routing.	Providers can dynamically serve genuine models for benchmarks and substitutes for regular traffic.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

- The Hardware-Based Solution:
  - Trusted Execution Environments (TEEs) offer secure, isolated enclaves on the hardware level.
  - Hardware-level attestation cryptographically guarantees that the correct model weights are loaded and the intended inference code is executing.
- Why TEEs Win:
  - Provable Integrity: Replaces probabilistic software guessing with mathematical guarantees.
  - Efficient: Introduces only modest performance overhead, unlike computationally prohibitive Zero-Knowledge Proofs (ZKPs).
  - Deployable: Practical implementation paths exist today (e.g., NVIDIA confidential computing on H100 GPUs).

Key Takeaways: Close the Verification Gap with Hardware Attestation

- 1. Model substitution is economically motivated and practically undetectable via software.
- 2. FP8 quantization barely affects benchmarks, enabling high-fidelity imitation.
- 3. Inference nondeterminism fundamentally kills log-probability fingerprinting.
- 4. Randomized substitution routes and benchmark evasion defeat statistical tests.
- 5. TEEs are the only currently viable path to verifiable LLM APIs.